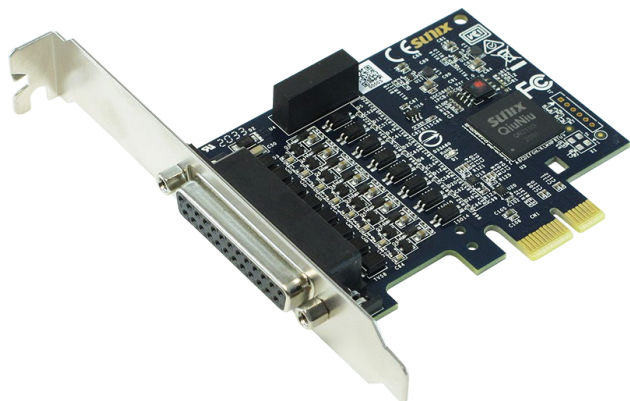




SDC0880I

PCI Express Industrial I/O Control Board

8-ch Isolated Digital Input

8-ch Isolated Digital Output



Introduction

SUNIX SDC0880I, industrial isolated digital I/O PCI-Express add-on card, is designed for PC-based IoT Gateway or desktop that enables data acquisition and I/O controls in industrial automation. This board built-in SUNIX Digital-I/O controller, **QiuNiu**, and also built with many of SUNIX advanced features and technologies. In addition, SUNIX provides API for users to program software under Windows and Linux operating systems. The software package includes dynamic library(DLL, SO), C , C# and Python programming sample codes, making it easier to develop application software. SUNIX SDC0880I board enriches digital I/O expandable capacity with cost-efficient design; it is the best digital I/O control solution for lite-industrial applications.

Features

- Designed to meet PCI Express Base Specification Revision 2.1.
- With high reliable SUNIX **QiuNiu** Digital-I/O controller.
- Plug-n-Play, I/O address and IRQ assigned by system.
- Ultra low power consumption (<1.5W) design for green environment and industrial application.
- High quality electronic components, low impedance and high stability ensure product quality and reliability.
- Support Microsoft Windows 10 and above with C/C++, C#, Python programming and provide sample code.
- Support Linux with C/C++, Python programming and provide sample code.
- Certified by CE, FCC, RoHS, and Microsoft approvals.
- Support both standard and Low profile bracket design to meet various of PC chassis.

Digital I/O Interfaces:

- Expands isolated eight (8) digital input channels with features for all channels.
 - Support both NPN & PNP devices
 - Support both dry and wet contact connections
 - With digital filter for noise reduction
 - Built-in 32-bit counter for all digital in channels
- Expands isolated eight (8) NPN type digital output channels with initial value (Booting & Restart) protection.
- Each channel built-in 2.5KV isolation protection for signal and power, allowing the input signals to be completely floated so as to prevent ground loops.

Specifications

Board Level

Interface	PCI Express
Controller	SUNIX QiuNiu Digital-I/O controller
BUS	PCI Express Gen2 x 1 (single Lane)
IRQ & IO	Assigned by System
ESD Protection	±15KV IEC61000-4-2 Air Gap Discharge ±8KV IEC61000-4-2 Contact Discharge

Digital Input

Type	NPN & PNP	Channel	8-channel
Dry Contact	Logic Level 0: Open; Logic Level 1: Close to GND	Wet Contact	Logic Level 0: 0 to 3VDC; Logic Level 1: 10 to 50VDC
Input Resistance	10KΩ	Isolation	2500 VDC
Counter	32-bit counter on all channel	Frequency	Input Range 2KHz max.
Digital Filter	Digital filter for noise reduction	Pin Define	DI1~DI8, COM, GND
Interrupt Mode	Event trigger (Rising Edge, Falling Edge, Both modes) & polling for all channels		
PCB Connector	DB25 Female connector		

Digital Output

Type	NPN	Channel	8-channel
Voltage Range	3.5 to 40VDC	Current	500mA per Channel
Isolation	2500 VDC	Pin Define	DO1~DO8, PWR, GND
Initial Protection	Load preset output value automatically during booting and restart		
PCB Connector	DB25 Female connector		

Software Support

OS	Windows 10 (x86/x64), Windows 11 (x64), Linux
API	Windows: DLL for Windows supporting C/C++, C# and Python. C, C#, Python sample code available Linux: SO for Linux supporting C/C++ and Python. C, Python sample code available

Declaration

Green Product	RoHS, WEEE
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Regulatory Approvals

Hardware	CE / FCC / VCCI / BSMI
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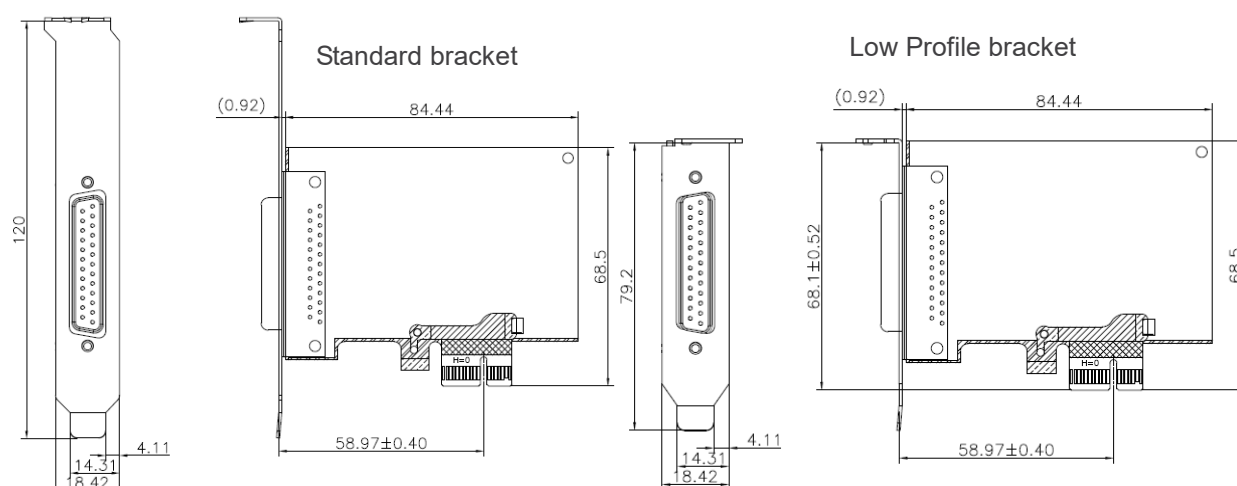
Environment

Operation Temperature	-40 to 85°C (-40 to 185°F)
Operation Humidity	5 to 95% (non-condensing)
Storage Temperature	-40 to 85°C (-40 to 185°F)

Dimension

PCB Dimension	84.44 x 68.5 mm
Bracket	Standard 121 mm / Low Profile 79.2mm
Bracket Space	1

Unit : mm

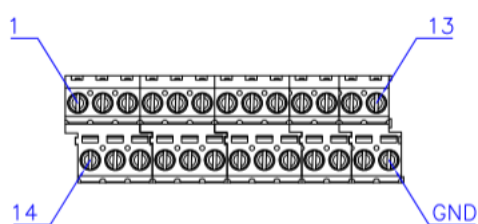


Package

- SDC0880I PCI Express Industrial I/O Control Board
- DB25 Male to Female connection cable, 100cm
- DB25 Male DIN-rail wiring board
- Low Profile Bracket
- Quick Installation Guide

Pin Assignment

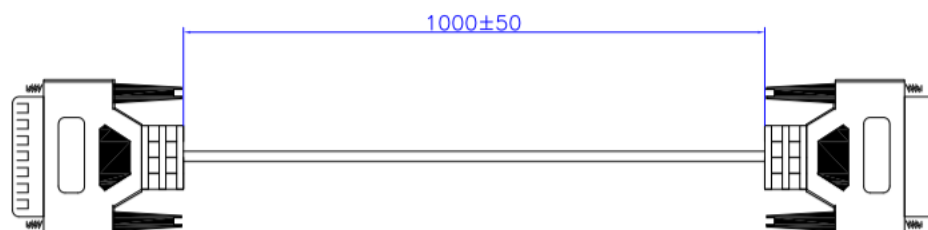
■ Digital Input / Output



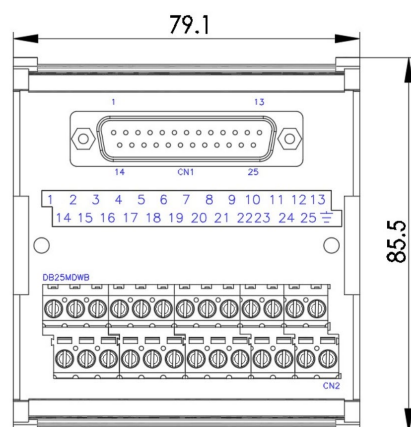
PIN	Digital Input	PIN	Digital Output
1	DI COM	14	DO PWR
2	DI COM	15	DO PWR
3	DI GND	16	DO GND
4	DI GND	17	DO GND
5	DI 1	18	DO 1
6	DI 2	19	DO 2
7	DI 3	20	DO 3
8	DI 4	21	DO 4
9	DI 5	22	DO 5
10	DI 6	23	DO 6
11	DI 7	24	DO 7
12	DI 8	25	DO 8
13	NC	GND	NC

Accessories

- DB25 Male to Female connection cable
- 100cm



- DB25 Male DIN-rail wiring board
- 79.1 x 85.5 mm



- Low Profile Bracket

